Masters Project Presentation

AGV Fleet VLA Intelligence

Multimodal Vision-Language-Action Decision Support for False-Positive Safety Stops in Autonomous Guided Vehicles

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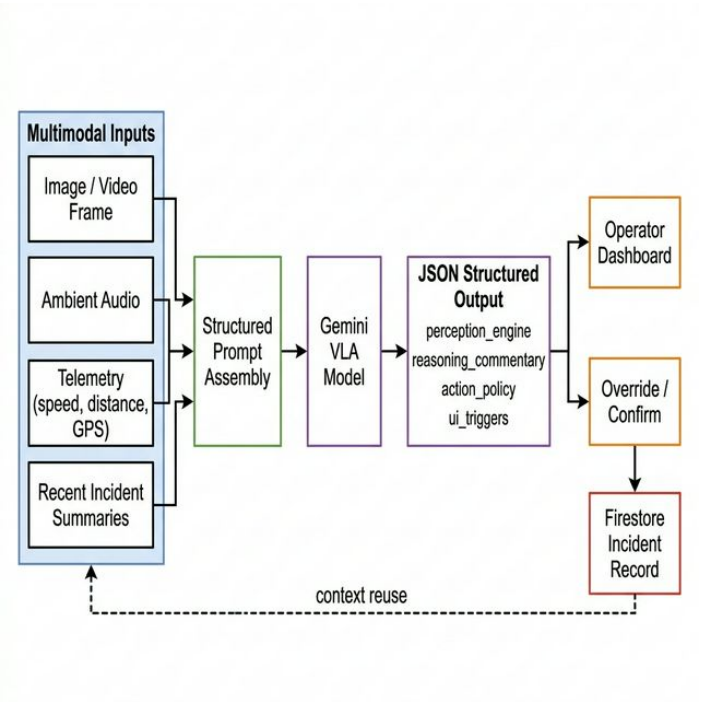


Human-in-the-loop advisory system for operator review of AGV stop events. Prototype evidence, not autonomous actuation.

Problem and Motivation

PROBLEM

- False-positive AGV stops reduce throughput and increase operator workload.
- Range sensors can detect distance, but not whether the obstacle is harmless or dangerous.
- Lightweight foil, wrapping film, paper, and soft barriers can trigger the same stop logic as real hazards.
- Operators need faster context to distinguish nuisance stops from genuine risk without removing human authority.

Benign nuisance

- Loose paper or wrapping material
- Reflective foil or soft plastic sheet
- Visually large but low-mass clutter

High-risk obstacles

- Dropped pallet or heavy cargo
- Human worker or moving object
- Solid equipment blocking the AGV path

Why this matters operationally

A warehouse operator must decide quickly whether the AGV is blocked by harmless material or by a genuine collision risk. The presentation therefore positions the project as decision support for remote assistance, not as autonomous control.

The presentation focuses on a narrow warehouse stop-resolution problem: improving operator review, not replacing the vehicle safety controller.

Objective, Scope, and Contribution

SCOPE

Objective

Build a multimodal decision-support dashboard that helps an operator review AGV stop events using media, telemetry, and recent incident

In Scope

- Operator-facing advisory dashboard
- Schema-constrained multimodal reasoning
- Incident persistence and traceability
- Prototype-level evaluation and audit tooling

Out of Scope

- Autonomous actuation or motion control
- Live fleet integration or teleoperation
- Industrial deployment readiness claims
- Large-scale validated benchmark claims

Implemented multimodal advisory prototype

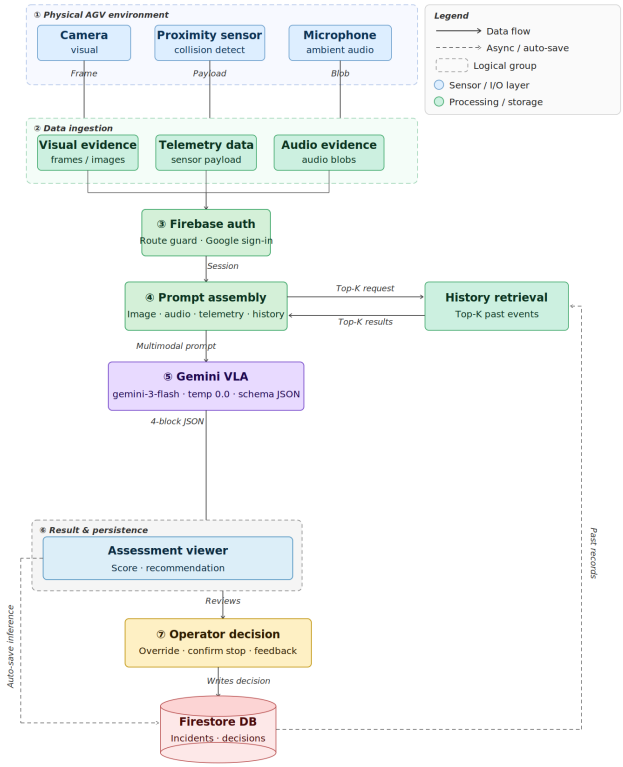
Structured operator workflow and persistence layer

Conservative evaluation and CSV-backed audit framework

System Architecture

SYSTEM

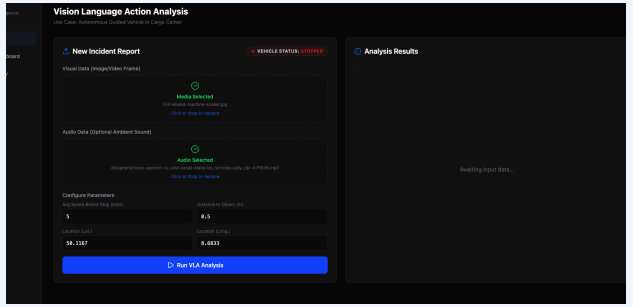
- Inputs: image or video frame, optional audio, typed telemetry, and recent incident summaries.
- Processing: Gemini multimodal analysis with a constrained JSON response contract.
- Outputs: danger score, recommended action, commentary, and UI trigger state.
- Persistence: Firestore stores both model output and operator decision for later audit and reuse.



Functional diagram of the advisory pipeline from multimodal inputs to operator-facing recommendation and persistence.

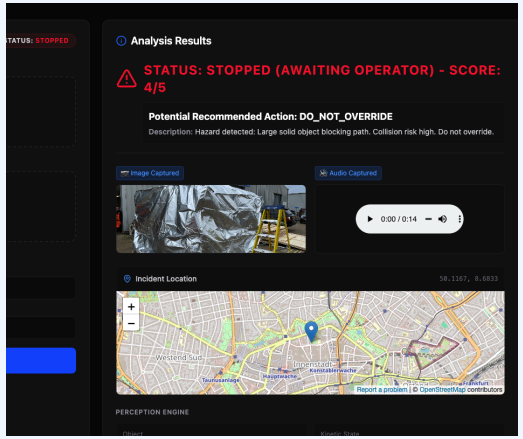
Prototype Workflow

WORKFLOW



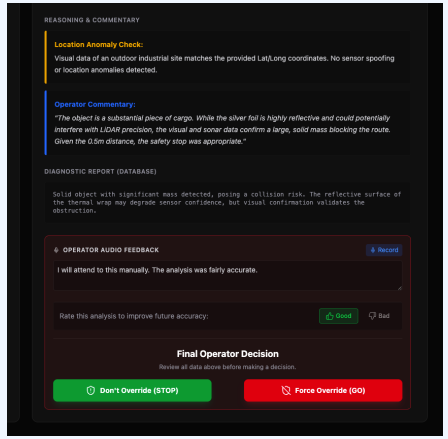
1. Capture

Operator provides incident media and typed telemetry in the prototype UI.



2. Analyze

The advisory engine returns status, score, and recommended operator action.



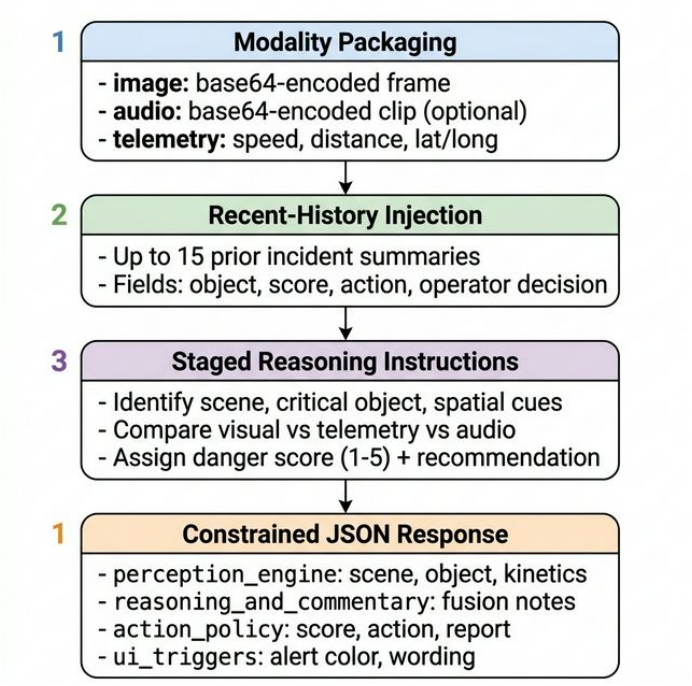
3. Decide

The operator confirms the stop or forces an override; the final action is logged.

The implemented workflow remains human-in-the-loop at the final decision point.

Model Contract and Decision Taxonomy

REASONING



Model configuration

- Main advisory model: gemini-3-flash-preview
- Deterministic decoding: temperature = 0.0
- Response format: application/json

perception_engine

reasoning_and_commentary

action_policy

ui_triggers

Danger-score mapping

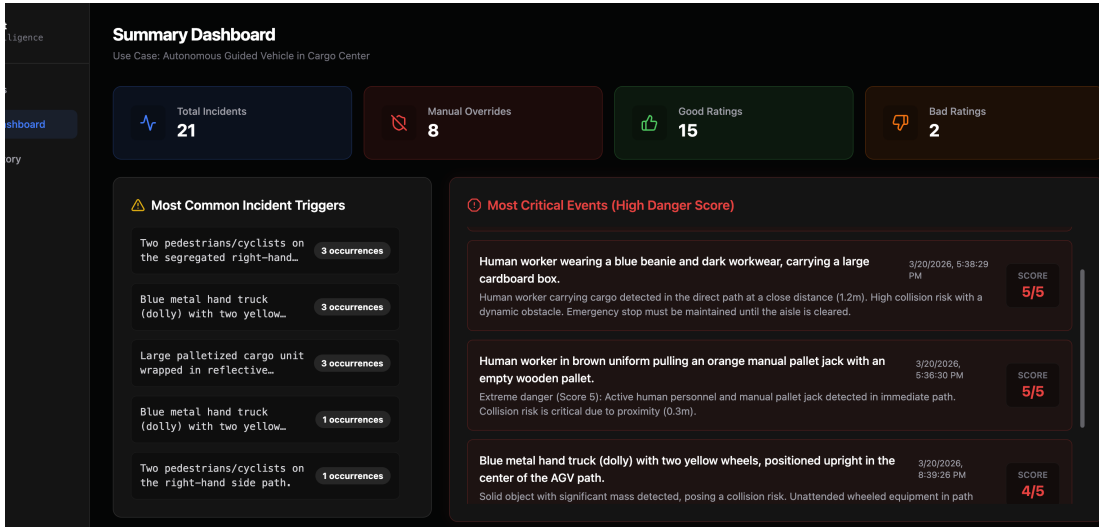
Scores 1-2: Safe to Override

Scores 3-5: Keep Stop / Do Not Override

Persistence, Dashboard, and Traceability

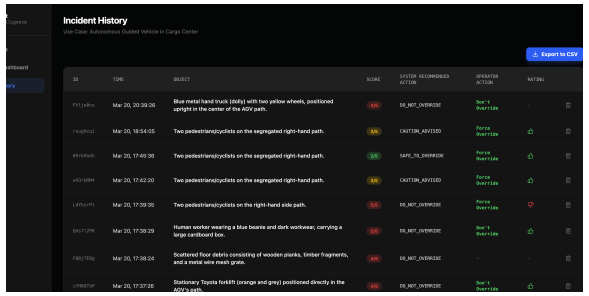
TRACEABILITY

- Firestore stores incident context, model output, operator action, ratings, and notes.
- Dashboard views aggregate total incidents, critical events, and low-score overrides.
- Incident history supports CSV export and later audit of system recommendations versus operator actions.
- History snapshots and source IDs are stored to make future consistency analysis traceable.



Traceability value

Persisted history snapshots and source IDs make future score-drift analysis auditable instead of anecdotal.



Evaluation Design

EVALUATION

Pilot benchmark

- 4 conservative text-surrogate scenarios
- 1 clearly safe, 1 borderline, 2 clearly unsafe
- 1 history-enabled case
- 0 repeated scenario pairs

Operational audit

- 20 exported dashboard incidents
- 3 repeated object groups and 9 repeated rows
- 1 repeated group with score variance
- 2 telemetry anomaly rows

Binary decision boundary review

Danger-score drift on repeated cases

Operator-action traceability

Data-quality anomaly detection

The current evidence is prototype-level and descriptive. It is not an industrial validation benchmark.

Operational Audit Results

RESULTS

Exported incidents	Repeated groups	Repeated rows	Variance groups	Telemetry anomalies
20	3	9	1	2

Interpretation

- The repeated-case evidence is small but useful for descriptive consistency checks.
- Only one repeated object cluster showed score variance in the export.
- Two repeated object groups remained stable at score 4 across all three occurrences.

Source: incident_history_audit.json generated from the exported dashboard CSV.

Repeated-Case Findings

FINDINGS

Alias	Score	Operator	Interpretation
PED-A	3/5	Force Override	Segregated lane; clean telemetry
PED-B	2/5	Force Override	Segregated lane; malformed lat/long
PED-C	3/5	Force Override	Segregated lane; clean telemetry

Stable groups

- Blue hand truck group: 4, 4, 4 and operator kept stop.
- Reflective silver cargo group: 4, 4, 4 across all three rows.

Conclusion: the observed 2 versus 3 split is localized and input-sensitive, not evidence of broad nondeterminism.

Challenges, Limitations, and Mitigations

LIMITS

Challenge	Effect on evidence	Mitigation in this project
Free-tier quota and API project limits	Model-side reruns could not be scaled credibly	Reported the offline audit truthfully and kept a quota-safe probe harness ready
Small pilot benchmark	No strong statistical performance claim is justified	Framed results as prototype evidence and audit evidence only
Manual telemetry entry in the prototype	Operator-entered inputs can introduce noise or malformed fields	Included telemetry anomalies explicitly in the operational audit
Text-surrogate benchmark inputs	Benchmark does not fully match raw-media production conditions	Separated the pilot benchmark from the operational dashboard audit
No formal HCI field study	Operator trust and speed effects remain unvalidated	Positioned HCI evaluation as future work

The advisory-only system boundary is intentional and remains appropriate given current VLM spatial-reasoning limits.

Conclusion and Future Work

OUTLOOK

Key takeaways

- A working multimodal AGV advisory prototype was implemented.
- The strongest contribution is the operator workflow, persistence layer, and traceability design.
- The present evidence is limited but honest, reproducible, and appropriate for a masters-level prototype.

Future work

- **Expand the benchmark from 4 scenarios to a wider and more varied**
- **Automate live telemetry ingestion through an API or streaming**
- **Move benchmark coverage from text surrogates to raw media**
- **Run an HCI study on operator trust, explanation quality, and review**

Best positioning: a credible academic prototype with a clear path to stronger evaluation.

Appendix: Selected References

Selected literature

- Ahn et al. (2022) - SayCan: grounded language for robotics.
- Brohan et al. (2023) - RT-2: vision-language-action modeling for robots.
- Tian et al. (2024) - DriveVLM: staged reasoning for autonomous driving.
- Parr et al. (2024) - remote monitoring, assistance, management, and driving.
- Guo et al. (2024) - SURDS: spatial reasoning limitations in multimodal models.

The full bibliography and methodological detail are documented in the project report.